

Amirreza Sokhankhosh

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HIGHLIGHT OF QUALIFICATIONS

- **Generative AI & LLMs:** Designed and deployed applications powered by LLMs using **LangChain**, **OpenAI API**, **vector databases (FAISS and Chroma)**, and **Ollama**, with expertise in **RAG**, fine-tuning (LoRA), and advanced prompting strategies (CoT, ReAct, Zero/Few-Shot).
- **Machine Learning Engineering:** Skilled in developing, training, and scaling ML models with **PyTorch**, **TensorFlow**, **Hugging Face Transformers**, applying best practices in distributed AI and model evaluation.
- **NLP & Computer Vision:** Applied deep learning to **text and image data** through projects in **natural language processing**, **OCR**, and **image detection/segmentation**, demonstrating the ability to handle multimodal ML challenges.
- **Full-Stack Development:** Proven track record in building and integrating end-to-end applications, combining **React (TypeScript)** for frontends with backends in **Flask**, **Django**, **NestJS**, and databases SQL and NoSQL such as **PostgreSQL** and **MongoDB**.
- **Cloud & DevOps:** Hands-on with **AWS**, **Docker**, **GitHub Actions**, and CI/CD pipelines to ensure reliable and scalable deployments.
- **Collaboration & Agile Practices:** Experienced in **Agile/Scrum** workflows, cross-functional collaboration, code reviews, and delivering high-quality, production-ready systems in both research and industry settings.

EXPERIENCES

University of Manitoba

May 2025 – Present

ML Engineer (Contract)

Winnipeg, Canada

- Preprocessed and normalized large **IoT datasets** in, producing model-ready inputs that stabilized convergence.
- Built a scalable ML pipeline with **PyTorch**, enabling secure, **parallel training** across 8+ distributed nodes with zero raw data exposure.
- Designed and deployed a Temporal Fusion Transformer for **multi-variate time-series forecasting**, reducing prediction errors by >20% for temperature, CO₂, and humidity data.
- Engineered a blockchain-integrated backend using Hyperledger Fabric and Express.js to automate model aggregation for 76 IoT sensor streams.

University of Manitoba

Sep 2023 – Jul 2025

Graduate Research Assistant

Winnipeg, Canada

- Designed **4** end-to-end distributed **ML pipelines** with **PyTorch** and **TensorFlow**, using modular code and containerization, supporting up to **36** training nodes.
- Built a microservice-based orchestration layer with **Flask** and **Express.js** and a fast **REST API** to coordinate distributed training reliably, achieving **90–100%** node uptime across all runs.
- Improved **model robustness** via adaptive validation schemes, handling inconsistent or adversarial updates, achieving **100%** detection of simulated adversaries.
- Optimized **training efficiency** by offloading heavy computation to servers, enabling lightweight clients, and reducing iteration time from **hours to minutes**.
- Co-authored **5+** **research papers** in distributed ML engineering, published/submitted to **top IEEE venues**.

Bobo App Ltd.

May 2024 – Aug 2024

Full Stack Developer Intern

Winnipeg, Canada

- Faced the challenge of improving customer support efficiency; collaborated with a team at **MILA** to develop an **intelligent chatbot** that automated routine customer interactions, reducing response times.
- Accelerated backend development by designing and implementing **RESTful APIs** with **PostgreSQL**, improving data retrieval performance and supporting scalable product features.

- Streamlined data management by automating CSV-to-SQL integration with Python scripts, reducing manual database population time by **80%**.
- Ensured smooth **full-stack integration** through cross-functional collaboration with the front-end team, aligning API contracts with user interface requirements to enhance product reliability.
- Enhanced team productivity within an **agile workflow** by using **Jira** and **Confluence** for task tracking, documentation, and sprint coordination.

Nadin Soft (Sadr Group Company)

Jul 2020 – Dec 2020

Full-stack Developer Intern

- Drove the successful completion of a critical internal project, addressing key deficiencies in API security, authentication, and data modeling for production deployment.
- Developed robust RESTful APIs with **Express.js**, implementing **JWT authentication** to secure user access and enable core application functionalities.
- Designed and implemented highly accurate MySQL database schemas for precise tracking and management of employee working and non-working hours, foundational for the project's business logic.
- **Deployed** fully functional features to the **live production server**.
- Contributed to an Agile Scrum development team, ensuring efficient progress and clear communication within sprints and **CI/CD pipelines**.

Aria

Sep 2025 – Oct 2025

Personal Project | [Code](#)

- Built an AI platform for generating tailored resumes and cover letters using **LangChain**, **React**, and **Flask API**.
- Implemented a **RAG** system over a curated knowledge base with OpenAI embeddings in **Chroma DB**.
- Added job caching with vector similarity in PostgreSQL + pgvector, reducing redundant LLM calls and latency.
- Developed a modular Flask backend with clean REST endpoints to orchestrate AI workflows and enable seamless front-end integration.
- Orchestrated LangChain + LangGraph workflows with stateful nodes for GPT-4 calls.

MarkMate

Jun 2025 – Aug 2025

Personal Project | [Code](#)

- Built an end-to-end AI grading platform using LangChain, React, and Django to auto-grade PDF submissions.
- Implemented a RAG pipeline to retrieve course materials and improve grading quality and consistency.
- Developed a modular Flask LLM microservice and orchestrated LangChain + LangGraph workflows.
- Engineered a robust Django backend with JWT authentication and REST APIs to facilitate seamless communication with the LLM service + a responsive React frontend for easy user interaction on all devices.

EDUCATION

University of Manitoba

Sep 2023 – Aug 2025

Master of Science in Computer Science (GPA: 4.4 / 4.5)

Winnipeg, Canada

- **Relevant Coursework:** Security & Privacy, Deep Generative Modeling, Blockchain & Distributed Systems: A+

K.N. Toosi University of Technology

Sep 2018 – Feb 2023

Bachelor of Science in Computer Engineering

TECHNICAL SKILLS

LLMs & NLP: LangChain, LangGraph, Transformers, Ollama

Machine Learning: TensorFlow, PyTorch, NumPy, Scikit-learn

Data Analysis & Visualization: Pandas, OpenCV, Matplotlib, Seaborn, Tableau

Programming Languages: Python, Java, JavaScript, TypeScript

Web Development: *Back-end:* Django, Express, Flask; *Front-end:* React, Material UI

Databases: PostgreSQL, MongoDB, MySQL, Redis, Db2, Supabase

Cloud & DevOps: Docker, Linux, CI/CD, Git/GitHub

Tools & Methodologies: Jira, Confluence, Swagger, Agile, Scrum